Do unique players pay off?

Speaker notes, November 6. Bold lines under Progress and Challenges are the slide; say the first point under each if time is short.

THE QUESTION WE ARE ANSWERING

Do unicorns pay off? Are they “steals” more often than typical prospects?

- **Unicorn:** a prospect whose size and speed are unusual for his position, on the athletic side. **Steal:** his career clearly beat what a pick at his draft slot usually returns. **Bust:** it fell well short.
- **Data, all from nflverse (2000–2026):** combine measurements to find the unicorns, the draft slot each one cost, and career value (Approximate Value) for what the team got back.
- **How we answer it:** compare unicorns with typical prospects on the six questions from our intro: what they cost (the price), whether their careers are better (unique or good?), whether they beat their slot (steal or bust?), where that differs (by round, by position), and a simple guide at the end (the payoff).
- **“Paying off”** means clearly more steals or fewer busts than typical prospects taken at the same slots.
- **So far:** the price has a first answer: unicorns cost more, 92% drafted at about pick 62 vs 66% at pick 108. Steal or bust is next; our first pass in October had them at 20% steals vs 17%, within noise.

HOURS LAST WEEK **~ 4 hours per group member.**

PROGRESS

Preprocessing, dealing with missing 40-times and other events, 24% of drills are missing

- **8,968** raw combine rows became **7,695** usable prospects: 27 classes, 11 position groups.
- **24%** of drill results are missing; the average prospect did 4.6 of the 6 drills. Only 5% of quarterbacks and 45% of receivers bench.
- No 40 time: almost none through 2016, **17%** in 2017–2021, **32%** in 2022–2026. Those 830 prospects are left out for now, nearly all for a missing 40.

Data inspection -> Eras matter, modern players are more athletic in general, positional differences, etc.

- 2022–2026 classes run the 40 about **0.6 SD faster** than 2000–2016 (roughly 0.07 s). Broad jump is up about 1.4 SD since 2000. Part may be selection: fewer prospects run now.
- **Chart:** weight and 40 time correlate **0.87** across positions but only 0.06 to 0.48 inside one. A 4.9 is slow for a receiver and fast for a defensive tackle.
- **Round:** career value falls from 44.7 in round 1 to 6.0 in round 7, so careers are judged against their slot.

Initial model work: Our 3 Unsupervised models are correlated but do not perfectly agree

- Rankings correlate **0.92 to 0.98**, yet only 241 of the 537 prospects flagged by any model are flagged by all three.
- **We now use one method, Mahalanobis distance.** It is closest to the three together (87% of the same flags) and has no settings or randomness. A new random seed changes 38 of Isolation Forest's 390 flags; a different neighbour count changes 53 of Local Outlier Factor's; Mahalanobis changes none.
- It flags **390** prospects (top 5% at each position). The **100** who are fast for their size are the unicorns, including Calvin Johnson, Tariq Woolen and Jordan Davis.

CHALLENGES

What to hold constant during analysis

- Comparing a prospect with his own era instead of all classes changes **1 in 5** flags (77 of 390). Without it, 2022–2026 prospects are flagged as fast 2.5× as often.

General trends change over-time

- **Unfinished careers:** classes through 2018 average about 18 career value per pick; 2022 is at 10.6 and 2025 at 2.3, so the newest classes cannot be graded yet.
- **Fewer prospects test at all, and top picks skip more:** since 2022, 36% of round 1–2 picks have no 40 time, vs 27% in rounds 5–7. That could tilt who gets flagged and how the groups compare, so we need to look into it.
- **The 2021 class** had no combine; its pro-day pool is flagged twice as often as other classes.

Specifying our definitions

- Unusual = top 5% at a position. Fast for size = 1 SD faster than size predicts; a cut-off of 0.75 gives 112 instead of 100.
- Steal and bust still need a cut-off we can defend: how far above or below the slot's usual return.

Why we are confident it can be answered: the data is already in hand. 23 classes (2000–2022) are old enough to grade, with 84 drafted unicorns against 4,333 typical picks. The method is settled, and a full first pass already ran end to end in October, so what is left is careful grading, not new data. The by-round and by-position answers will rest on smaller groups (30 first-round unicorns).

Next: add the other five drills, then grade careers against the draft slot.

If asked

Backup answers for questions after the update.

What is an outlier, and what is a unicorn?

- **Outlier:** top 5% at his position by Mahalanobis distance, meaning far from the centre of his position on height, weight and speed for size.
- **We choose** the inputs, the comparison group (same position, classes within 2 years) and the 5% line. The method only ranks.
- It cannot tell good from bad, so a second rule splits the flagged prospects by direction:

Flagged group	Prospects	Drafted	Pick
Fast for size (unicorns)	100	92%	61.5
Unusual build	131	64%	118.5
Slow for size	159	23%	191.5
Typical (not flagged)	7,305	66%	108

Pick = median overall pick among those drafted. Draft figures cover classes through 2025.

Why one method, and why Mahalanobis distance?

- We were told to use one method. The three rank prospects almost identically (0.92 to 0.98), so little is lost by keeping one.
- **Closest to all three together:** it shares 87% of its flags with the three-model average (Isolation Forest 82%, Local Outlier Factor 85%).
- **Stable:** no settings and no randomness, so the same flags on every run. A new random seed changes 38 of Isolation Forest's 390 flags; 20 neighbours instead of 35 changes 53 of Local Outlier Factor's.
- **Easy to explain:** how far a prospect is from the centre of his position, in standard deviations, given how height, weight and speed go together. The 5% line sits near 2.8; Jordan Davis (341 lb, ran 4.78) is at 4.4.
- **The two we dropped:** Isolation Forest counts how few random yes/no cuts isolate a player; Local Outlier Factor asks whether his neighbourhood is emptier than his neighbours'.

What about strength (bench press)?

- **Not measured yet.** "Athletic" so far means the 40 time relative to size; bench is one of the five drills still to add.
- **Size is not the obstacle:** within a position, height and weight explain only 2–17% of bench reps (0–23% for the 40), so "strength for size" can be built the same way as speed for size.
- **Missing data is:** 64% have a bench result (quarterbacks 5%, receivers 45%), and only 15–36% per class since 2022, down from 72–83% in 2018–2021. First-rounders skip it more: 61% benched vs 71–75% in later rounds.
- **What it measures:** reps at 225 lb reward endurance and short arms more than peak strength, and arm length is not in our data.
- **Plan:** a fourth feature in a secondary run (non-quarterbacks, 2000–2021 classes), with no filling-in of skipped benches. The jumps are better covered: vertical 83%, broad jump 82%.

What is new since the intro deck?

- **Cleaning, step by step:** 8,968 rows to 7,695, with 11 wrongly attached draft picks fixed.
- **The missing-data picture:** drills and 40 times.
- **Eras matter:** recent classes test better.
- **Position differences:** the chart.
- **Three models compared, one chosen.**
- **The era comparison,** now built into the model.
- **Why strength is not in yet.**

Intro deck numbers, re-checked

	Intro deck	Today's notebook
Prospects	7,690	7,695
Flagged as unusual	390	390
Unicorns	108	100
Share drafted	88% vs 67%	92% vs 66%
Median pick	66 vs 109	61.5 vs 108
Round-one steal rate	37% vs 18%	not redone yet

The numbers moved because of the cleaning fix, and because today's model uses one method on 3 measurements instead of three methods on 8. The takeaway is the same.

Numbers come from initial_analysis.ipynb as run on November 6, 2026. The bench-press figures, the testing-by-round figures and the counts of gradeable picks were computed separately from the same cleaned data. The October steal rates and the intro deck numbers come from the earlier prototype and have not been re-checked.